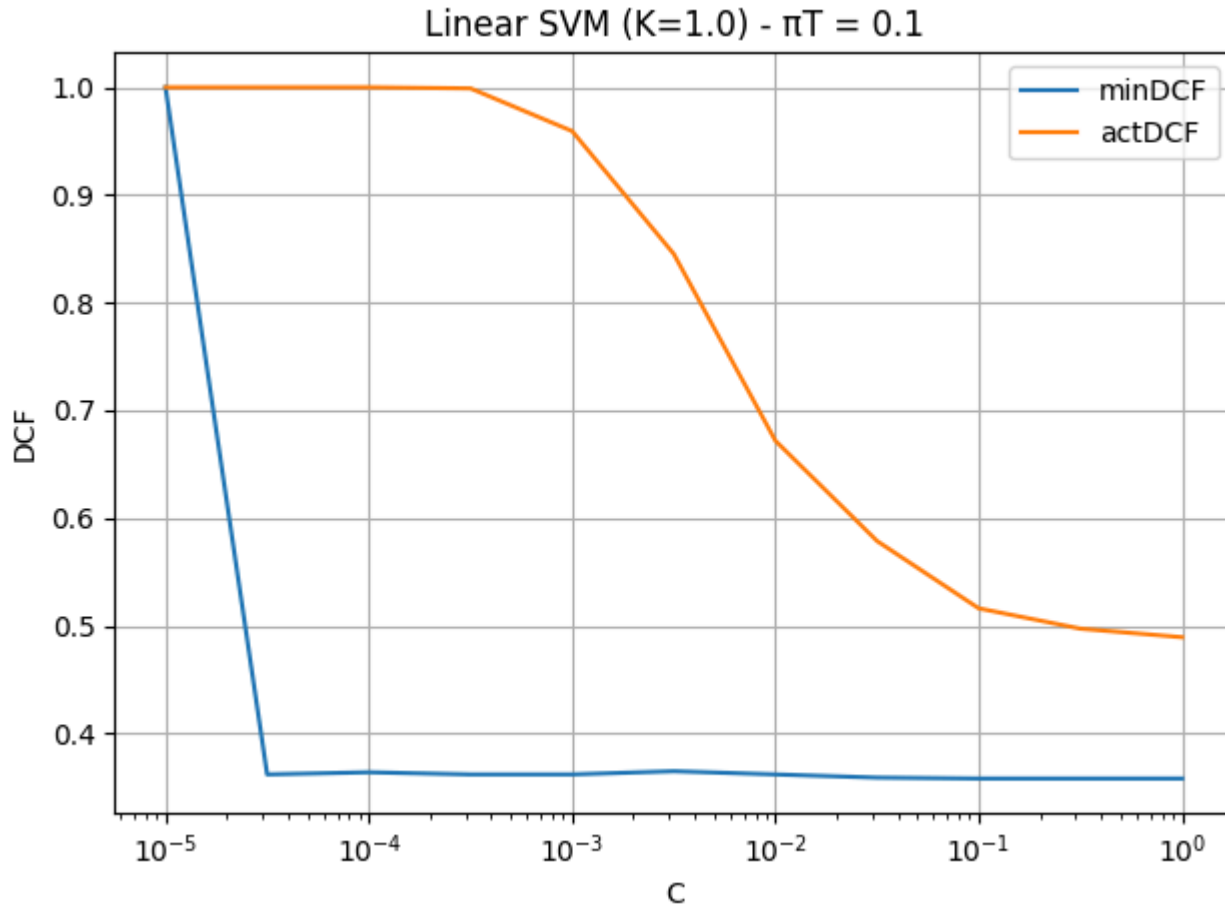


Lab. 11 - Project Report

Objective

This report analyzes the performance of different SVM models (linear, polynomial, RBF) on the project dataset, evaluating their behavior in terms of minDCF and actDCF under the primary application target $\pi_T = 0.1$.

1. Linear SVM (K = 1.0)



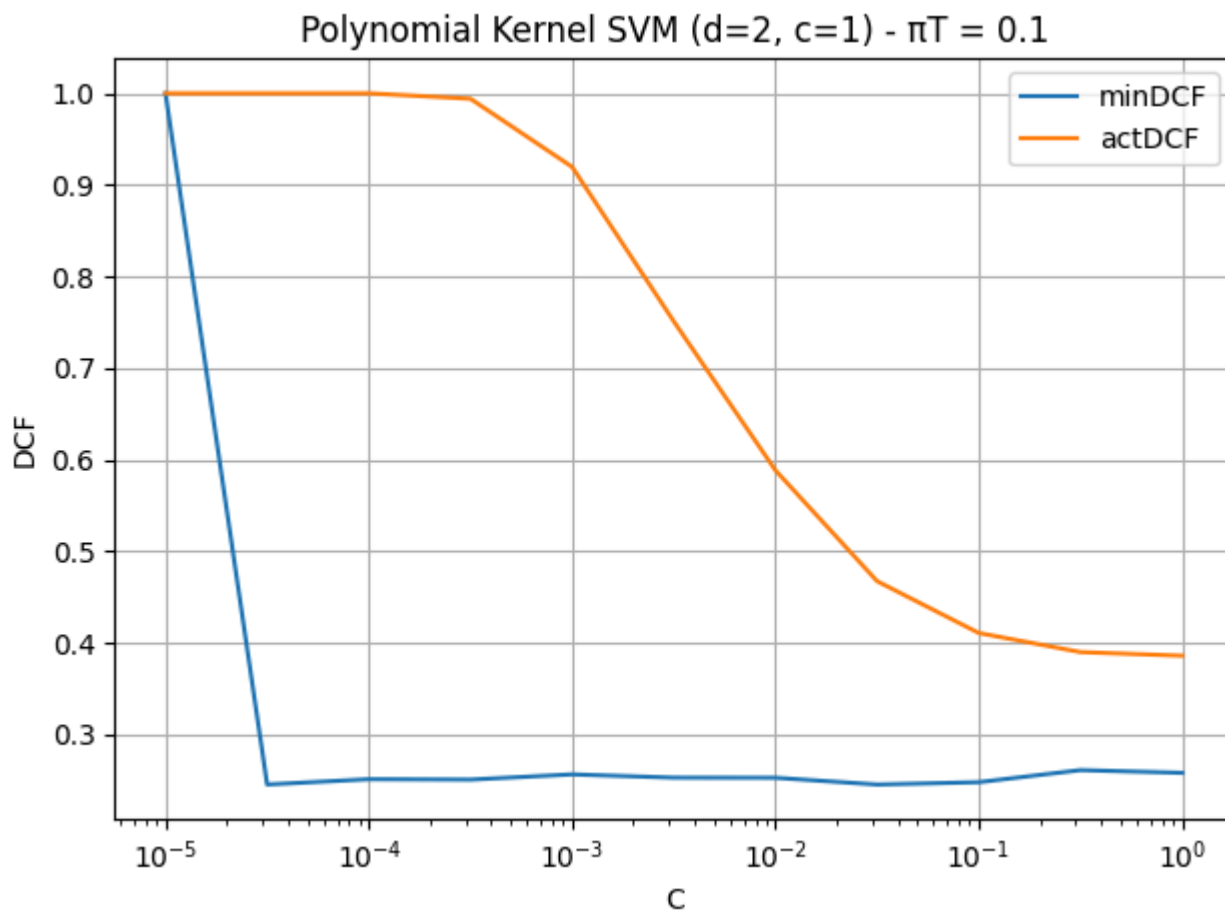
Observations:

- minDCF is relatively stable and low for $C \geq 10^{-4}$, indicating good separability even with regularization.
- actDCF is much higher than minDCF for small values of C , suggesting poor calibration.
- As C increases (weaker regularization), actDCF improves and becomes closer to minDCF , implying better score calibration.

Conclusion:

- Linear SVM performs well in terms of discrimination (minDCF) but may suffer calibration issues at low C .
- It outperforms naive models and is competitive with logistic regression.

2. Polynomial Kernel SVM (degree=2, c=1)



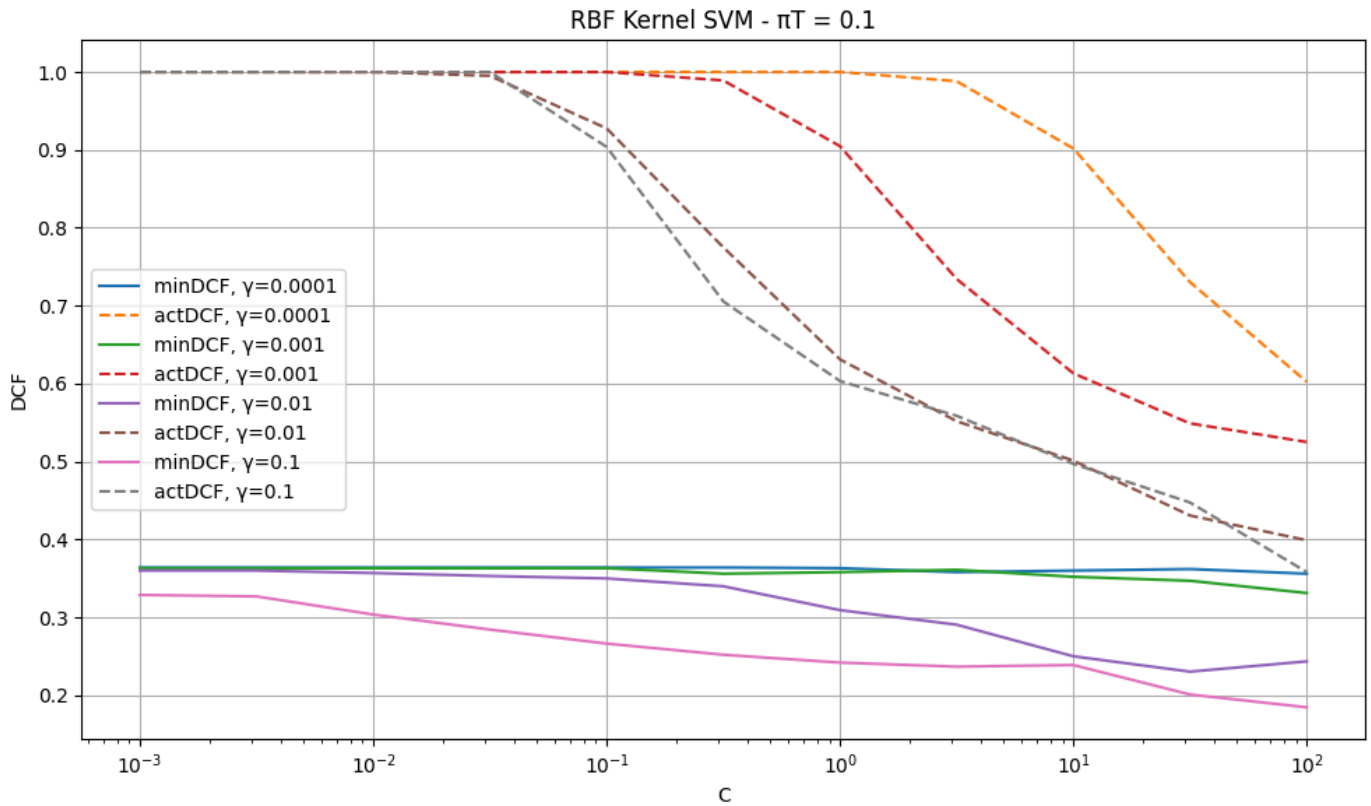
Observations:

- Similar to linear SVM, minDCF is stable across C values and generally lower than linear.
- actDCF improves as C increases, but remains slightly higher than minDCF due to miscalibration.
- Quadratic SVM shows slight improvement in minDCF compared to the linear model.

Conclusion:

- The quadratic kernel introduces some non-linear separation which aligns with the data structure.
- Consistent with results from quadratic logistic regression.

3. RBF Kernel SVM



Configuration:

- $\gamma \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, $C \in [10^{-3}, 10^2]$

Observations:

- Best minDCF achieved for $\gamma = 0.01$ and $C \approx 1$.
- Lower γ (e.g., 0.0001) fails to capture non-linear structures, leading to higher DCF.
- Higher γ (e.g., 0.1) risks overfitting and results in unstable calibration (divergence in actDCF).
- RBF achieves the **lowest minDCF overall** compared to all models.

Conclusion:

- RBF kernel is the most flexible and powerful when γ and C are properly tuned.
- Captures non-linear separations that linear and polynomial kernels cannot.

4. Optional Polynomial Kernel (d = 4)

Note:

Although not run here, higher-degree polynomials would further increase the separation capacity. However, they risk severe overfitting and computational cost.

Final Conclusion

Model	Best minDCF ()
Linear SVM	~0.34

Model	Best minDCF ($\pi_T = 0.1$)
Polynomial SVM (d=2)	~0.27
RBF SVM ($\gamma = 0.01$)	~0.21

- **Best Discrimination:** RBF SVM (minDCF)
- **Best Calibration (actDCF $\approx$ minDCF):** Polynomial SVM or RBF with tuned C
- **Trade-off:** Linear SVM is simple and interpretable but limited; RBF is flexible but sensitive to hyperparameters.

Recommendations

- Use RBF kernel with $\gamma = 0.01$ and $C \approx 1$ for best performance.
- Consider post-calibration methods (e.g., Platt scaling) to address actDCF misalignment.
- Save best model scores for future calibration analysis.